

Full-duplex visible light communication system based on single blue mini-LED acting as transmitter and photodetector simultaneously.

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Abstract—GaN-based light emitting diodes (LEDs) have potential as photodetectors (PDs) for visible light communication (VLC). In current studies on LED-based PDs, devices are operated in a reverse-biased or zero-biased state. In this work, the performance of an $80 \times 120 \mu\text{m}^2$ blue mini-LED as both a PD and light source was studied. An aluminum oxide passivation was deposited on the sidewalls of the active region using atomic-layer deposition. In addition to improving the external quantum efficiency of mini-LEDs and reducing leakage current, sidewall passivation can also increase the responsivity of mini-LED PD. At the injection current used for display, the mini-LED device still worked as a PD with a good response to illumination. At a bias of 3 V (corresponding to injection current of 11.5 mA), the mini-LED PD had a response of 0.243 A/W and a high on-off ratio of 1.33×10^7 . A cascaded bridged-T amplitude equalizer was used to extend the OE -3 dB bandwidth of mini-LED PD from 16 MHz to 116 MHz at 3V bias. Based on the system above, a VLC link over 1 m with a data rate of 528 Mbps is achieved using 16-ary quadrature amplitude modulation orthogonal frequency division multiplexing. Moreover, the mini-LED PD could simultaneously act as an uplink detector and a downlink transmitter in the same full-duplex VLC system. With transmission distance of 1m/ 1m, a transmission rate of 350 Mbps in the uplink and downlink was achieved. With transmission distance increasing to 1.2 m/1.2 m and 1.5 m/1.5 m, full-duplex VLC systems are also capable of achieving transmission rates of 300 Mbps and 225 Mbps, respectively. The proposed method allows the same mini-LED to be used simultaneously as a display and VLC transmitter and receiver,

which not only simplifies the structure of the full-duplex communication system and reduces system power consumption, but also facilitates the integration of multifunctional small LED devices.

Index Terms—Visible light communication, light emitting diodes, detector, optoelectronic devices.

I. INTRODUCTION

Gallium nitride (GaN)-based light-emitting diodes (LEDs) are considered the most promising next-generation display technologies [1, 2]. Micrometer-level-LED with a size less than $200 \mu\text{m}$ have the advantages of high brightness, power saving, high color saturation, and high reliability over traditional display technologies, such as liquid crystal displays (LCDs) and organic LEDs (OLEDs) [3-6]. GaN-based LEDs with small sizes can satisfy the demands of sophisticated devices, including smart watches, virtual reality/augmented reality (VR/AR), microprojectors, and ultra-high-definition TVs, which enables the development of mini-LEDs and micro-LEDs [7].

In addition to displays, GaN-based LEDs also play an important role in visible light communication (VLC) [8-12]. VLC has garnered significant attention owing to its advantages of low power consumption, no electromagnetic interference, high transmission rate, and power saving, making it a promising candidate for wireless communication systems. By minimizing the size of the device to the micron-scale, mini/micro-LEDs can achieve lower capacitance and higher modulation bandwidth, enabling LEDs to transmit at high data rates in VLC applications, which is highly promising for such applications [8, 13, 14]. Due to its unique properties, GaN-based LEDs have the potential to enable both display and communication. However, the current density of micro-sized LEDs as transmitters in VLC systems usually reaches several kA/cm^2 , which increases the power consumption [15]. With the growing demand for IoT featuring low power consumption and miniaturization, short-range VLC systems with low power consumption are beginning to attract the attention of researchers [16, 17]. Zhu et al. achieved Gbps-level transmission rates at distances of

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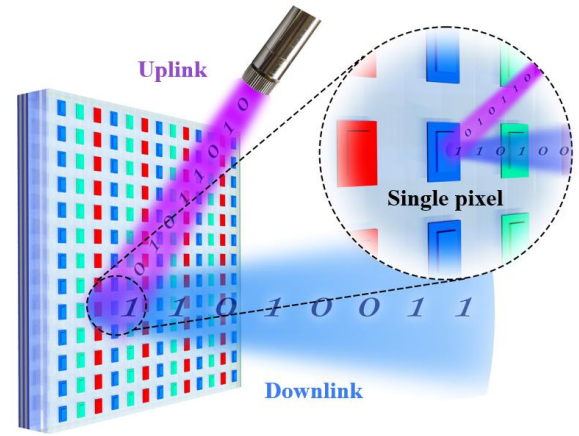
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0.2 m and 0.5 m based on nonpolar micro-LEDs at low injections of 50 A/cm² and 320 A/cm², respectively [18].

In addition to serving as VLC transmitters, GaN-based LEDs also have potential as photodetectors (PDs). Most commercial PDs are based on Si or GaAs, and the receiving spectrum covers a wide range, which causes significant optical noise [19, 20]. GaN-based LEDs have wavelength selectivity when used as PDs and can only receive light waves with wavelengths lower than their own; that is, they have natural filter characteristics [21]. Furthermore, compared with other PDs, mini/micro-LED PDs have the advantages of small size, low cost, easy integration, and mass production.

Based on the advantages of mini-LED as display, VLC transmitter and receiver, we can imagine the application scenario shown in Figure 1, the mini-LED display is displaying HD video. At the same time, if the pixels in the display can simultaneously perform data transmission and detection and achieve full-duplex communication at relatively low operating current density sufficient for display, it is expected that a low-power multifunctional information display can be realized without additional network modules. Each pixel in this concept of mini-LED based information display can simultaneously display, transmit and receive signals, which not only simplifies the system architecture and reduces system power consumption, but also facilitates the integration of multifunctional mini-LED devices. Such multifunctional information display has a wide range of promising applications in devices that require high-speed wireless networks and desire to reduce power consumption, such as VR/AR devices and outdoor large-screen displays. Furthermore, it also has potential to be used as an information display chip for miniaturized IoT applications to enable display applications combined with low-power and energy-efficient VLC, such as small-sized smart wearable devices. Such applications have lower transmission distance requirements. Therefore, the short-range VLC with moderate power consumption is more suitable for these applications [22]. However, the signal detection performance of mini-LED under forward bias sufficient for display or data transmission are still poorly studied.



Mini-LEDs based displays

Fig. 1. Concept of multifunctional mini-LEDs based information display.

In recent years, LEDs have been gradually explored as the detectors at the receiving end of VLC systems. We summarize the VLC systems using LEDs as detectors reported in the literature in table 1. In 2015, Stepniak et al. implemented a red LED-to-LED VLC link at a distance of 5 cm with a data rate of 110 Mbps, in which the LED detector is biased at -30 V [23]. With the advancement of micro-sized LED production technology, micro-sized LEDs as receivers have been manufactured and used in high-speed VLC systems. In 2019, Liu et al. realized a 185-Mbps/1-m unidirectional VLC system based on μ LED arrays as detectors at 0 V bias [24]. In 2020, Mariam et al. proposed a unidirectional 200 kbps/7 cm communication system based on red-green-blue (RGB) LEDs, of which the red sub-LED is employed as photodetector while the green sub-LED is the transmitter [25]. In 2020, Jung et al. demonstrated 49 kbps/0.15 m full duplex VLC system based on two different yellow LEDs at -5 V bias as uplink and downlink receivers, respectively [26]. In 2021, Lin et al. achieve a 2.3 m full-duplex UWOC system with a data rate of 60 Mbps based on a green μ LED array as the transmitter and another μ LED array at -5 V as the receiver. [27]. In 2021, Jung et al. implemented a 3×3 multiple-input multiple-output (MIMO) LED-to-LED communication system using RGB colors at a data rate of 30.62 kbps over a 10 cm transmission distance [28].

Table 1. Comparison of VLC systems based on LED detection

Year	Transmitter/Receiver	Bias (V)	Modulation format	Distance (m)	Data rate (Mbps)	Unidirectional/ Bidirectional	Ref
2015	Red LED/Red LED	-30	PAM-2	0.01	100	Unidirectional	23
2019	405 nm LD/Blue μ LED array	0	OOK	1	185	Unidirectional	24
2020	Green LED/Red LED	0	OOK	0.07	0.2	Unidirectional	25
2020	Green LED-1/Yellow LED-1; Green LED-2/Yellow LED-2	-5	32-QAM-OFDM	0.14/0.14	0.049	Bidirectional	26
2021	Blue LD/Green μ LEDs array-1; Green μ LEDs array -2/APD	-5	OOK	2.3/2.3	60	Bidirectional	27
2021	RGB-LEDs/ RGB-LEDs	0	32-QAM-OFDM	0.15	0.03	Unidirectional	28
2022	405 nm LD/blue mini-LED; Same blue mini-LED/APD	3	16-QAM-OFDM	1/1	350	Bidirectional	This work

As shown in table 1, in past studies on LED-based PDs, devices are operated in a reverse-biased or zero-biased state. In this operating state, the LED is turned off and cannot be used for display or VLC transmitter, with a single function of signal detection. Even if it is envisaged that part of the LEDs in array are used for displays or VLC transmitters (Tx) while the remainder of the LEDs operate as PDs, this requires complex drive circuit control and will reduce effectiveness. However, its signal detection performance under forward bias sufficient for display or data transmission and applications are still poorly studied. If commercial mini/micro-LEDs can achieve signal detection while acting as displays and transmitters in VLC system simultaneously, it will play a role in promoting the realization of multifunctional mini/micro-LEDs. This also provides the possibility for the device to operate simultaneously as a transmitter and receiver for high-speed VLC.

Based on the considerations discussed above, in this study, we demonstrate another advantage of commercial mini-LED as an information display, that is, single mini-LED can simultaneously achieve signal transmission and detection while working as a display. An aluminum oxide (Al_2O_3) passivation was deposited on the sidewalls of the active region using atomic-layer deposition (ALD). In addition to improving the external quantum efficiency (EQE) and reducing leakage current, sidewall passivation can also increase the responsivity of mini-LED PD. This will further improve the performance of mini-LED PD when emitting light. At a forward bias voltage of 3 V, the device has an injection current density of 120 A/cm^2 , which is sufficient for illumination or display, it still had a considerable detection responsivity of 0.243 A/W . With the mini-LED with size of $80 \times 120 \mu\text{m}^2$ acting as PD, a VLC link over 1 m with a data rate of 528 Mbps is achieved using 16-ary quadrature amplitude modulation (16 QAM) orthogonal frequency division multiplexing (OFDM). This indicates that the mini-LED could still act as PD for VLC, even at an injection current sufficient for display. In addition, based on this experimental result, we further designed a full-duplex communication system in which a single mini-LED biased at 3 V acted as both an uplink detector and a downlink transmitter, together with a 405 nm laser diode (LD) and an avalanche photodiode (APD). With transmission distance of 1m/ 1m, a transmission rate of 350 Mbps for the uplink and downlink was achieved, with bit error rates (BERs) of 2.1×10^{-3} and 2.8×10^{-3} , respectively, which can realize the function similar to the antenna in wireless communication. With transmission distance increasing to 1.2 m/1.2 m and 1.5 m/1.5 m, full-duplex VLC systems are also capable of achieving transmission rates of 300 Mbps and 225 Mbps, respectively. The proposed method allows the same mini-LED to be used simultaneously as a display, VLC Tx, and VLC Rx at relatively low operating current density, which simplifies the structure of the full-duplex communication system, reduces system power consumption, facilitates the integration of multifunctional small LED

devices, and verifies the potential of GaN-based LED applications in the field of display and full-duplex VLC.

II. DEVICE STRUCTURE AND EXPERIMENT

A. Device structure

Figure 2 (a) illustrates the cross section of the proposed blue mini-LED with size of $80 \times 120 \mu\text{m}^2$ acting as a PD. These samples were grown on a (0001) *c*-plane patterned sapphire substrate via metal-organic chemical vapor deposition. Subsequently, n-type GaN and InGaN/GaN MQWs and a p-type GaN layer were deposited. A 200 nm-thick layer of indium tin oxide (ITO) was deposited on top of the epitaxial structure. Then, hydrogen chloride (HCl) and ICP-RIE were used to etch the ITO film and mesa. Titanium/aluminum/nickel/gold (Ti/Al/Ni/Au) were deposited as electrodes. Moreover, atomic layer deposition (ALD) was used to grow a 100 nm-thick Al_2O_3 passivation and 50 nm-thick silicon dioxide (SiO_2) layer using trimethylaluminum and H_2O , with Ar purging at 300°C . The distributed bragg reflectors (DBRs) were deposited by plasma enhanced chemical vapor deposition (PECVD). We then flipped the processed LED chip onto the submount. Thermal-sonic process was then performed to complete FC bonding. The specific fabrication steps can refer to previous work [29]. This commercial blue mini-LED was designed for light emission, and ALD was introduced to eliminate surface defects caused by the size effect and reduce leakage current [30]. Chen et al. also demonstrated that ALD sidewall passivation reduces the trapping of photogenerated carriers by the sidewall defects of the device, thereby further improving the photocurrent of the LED [31]. Therefore, ALD sidewall passivation will effectively improve the performance of mini-LEDs as transmitter and receiver simultaneously in VLC systems. It is worth noting that we used a single mini-LED instead of a mini-LED array for experimental tests to prevent other devices in the array from affecting the LED's emission performance when receiving signals in the designed full-duplex communication system. Figure 2(b) shows the electroluminescence (EL) spectra of the blue mini-LED for different injection currents, ranging from 1 to 12 mA.

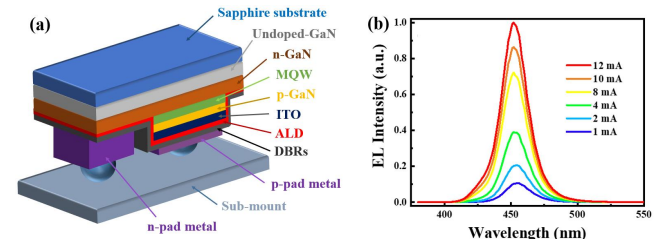


Fig. 2. (a) Cross section of proposed blue mini-LED acting as PD. (b) EL spectra for different injection currents.

B. I-V Curve and External Quantum Efficiency

To demonstrate the improvement of sidewall passivation on EQE, leakage current and responsivity, the standard sample with the same sizes and without passivation layer were also studied as comparison. As the epitaxial structure and processing progress of samples are consistent, ALD treatment is an important factor that causes differences in

1 electroluminescence performance characteristics. Figure 3
2 (a) show the current versus voltage (I-V) curve of devices.
3 Since the ALD passivation layer repairs the leakage channel
4 caused by the non-radiative recombination on the sidewalls,
5 the samples of this work have lower leakage current
6 densities (about 10^{-10} A) than the standard samples without
7 sidewall passivation [32]. Sidewall passivation can also
8 reduce the efficiency drop caused by SRH non-radiative
9 recombination caused by sidewall defects [33]. The peak
10 EQE of sample with ALD is 44.7%, which is 10.6% higher
11 than that of the standard sample, as shown in Fig. 3 (b).

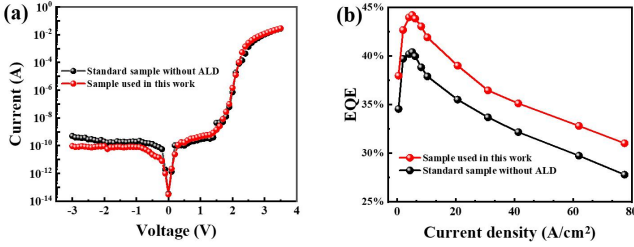


Fig. 3. (a) I-V curve and (b) EQEs of the sample used in this work and standard sample without ALD.

C. Detection Performance of Mini-LED PD When Turned on

The optoelectronic performances of the mini-LED device used in this work and the standard sample as a PD were measured. A 405 nm laser diode (LD) was driven as the emitter with different DC currents. The optical signals were then received by the mini-LED PD after a 1 m free-space transmission distance. A pair of lenses was added in the light path to focus the light onto the mini-LED PD. A bias voltage was applied to the mini-LED PD using another current source. During the experiment, we used a power meter (Thorlabs PM100D) to measure the optical power received by the PD.

I-V curves for different optical powers received from the LD were measured to verify the optoelectronic performance of the mini-LED in this work and standard samples, as shown in Fig. 4(a) and (b). The received optical power is the incident-light power on the active area of the mini-LED PD, which is converted using the following:

$$P_{in} = \frac{P_{LD}}{A_{LD}} \times A_{LED} \quad (1)$$

where P_{LD} represents the light output power of the LD, and A_{LD} represents the area of the light spot for the 405 nm LD focused on the mini-LED PD, which was measured to be approximately $430 \mu\text{m}^2$ using a microscope. A_{LED} represents the area of the mini-LED PD.

The current of the LED under dark conditions is given by the following formula:

$$I_f = \exp\left(\frac{qV}{\eta kT}\right) \quad (2)$$

where I_f is the forward current density, V is the applied forward bias, q is the electronic charge, k is the Boltzmann constant, T is the temperature, and η is the ideal factor.

In the case of reverse bias and a small injection current, as the mini-LED PD receives optical power, the current decreases because the photogenerated carriers generate a

reverse photocurrent under the action of the built-in electric field; therefore, the total current flowing in the outer circuit can be expressed as

$$I = I_f - I_{ph} \quad (3)$$

where I_{ph} is the photocurrent generated in the mini-LED PD. We observed that the current through the mini-LED began to increase with an increase in the received optical power at biases larger than 2.5 V, which suggests that the applied electric field provided by the bias at this time was larger than the built-in electric field.

The insets of Fig. 4(a) and (b) demonstrate that the photocurrent still increases with the received optical power at turned-on bias voltage, which means that even when emitting light under current injection, the devices could still act as a detector with the direct-current (DC) filtered out. In addition, it can be seen from the figure that the photocurrent produced by the samples with ALD sidewall passivation is higher than produced by the standard sample at different bias voltages. Based on the interpretation of trap defects in the passivation interface, laser radiation excites the InGaN/GaN active region to generate photo-generated carriers, which drift under the action of the built-in electric field to generate photocurrent. However, parts of the carriers are trapped by the sidewall traps during the drift process, resulting in a drop in photocurrent.

The responsivity (R) can describe the photoelectric conversion ability of the PD and is given by the following formula [24]:

$$R = \frac{I_{ph}}{P_{in}} \quad (4)$$

Sidewall defects lead to a drop in the responsivity of mini-LEDs as PDs due to the reduction in photocurrent. We calculated the responsivity of the mini-LED PDs at different bias voltages, as shown in Fig. 4(c). The responsivity of the devices used in this work was approximately 0.712 A/W when the bias voltage was -5 V, which is higher than 0.546 A/W of the standard sample without sidewall passivation. As the bias voltage increased, the responsivity of the mini-LED PD gradually decreased. This is because the built-in electric field was canceled by the applied electric field, resulting in the formation of a smaller photocurrent by the photogenerated electron-hole pair. After mini-LED PDs were turned on, the responsivity reaches the lowest value at bias of 2.5V, and after that the responsivity of the mini-LED PD gradually increases with the bias voltage increases. We think that it is due to the fact that the external bias voltage is already larger than the built-in electric field after 2.5V, resulting in the photocurrent increases with the external bias voltage. When the bias voltage is increased to 3 V, the responsivity of the device used in this work is 0.243 A/W, which is still higher than 0.209 A/W of the standard sample. This indicates that the mini-LED PD can still generate significant photocurrent under illumination even at injection currents sufficient for display. Therefore, ALD sidewall passivation will effectively improve the performance of mini-LEDs as transmitter and receiver simultaneously in VLC systems. Therefore, ALD sidewall passivation will

- 1 improve the optoelectronic performance of mini-LEDs
- 2 working as displays, emitters, and photodetectors.

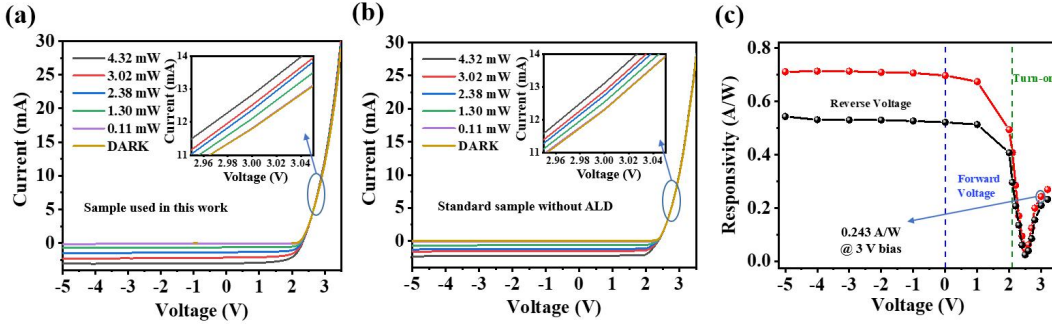


Fig. 4. Current–voltage (I–V) curves of (a) the sample used in this work and (b) standard sample without ALD under dark and illumination conditions with different received optical powers. (c) The Responsivities under varying received optical powers at bias voltages of –5 to 3 V for samples.

Based on the better performance, we performed experiments with samples with ALD sidewall passivation and further measured the properties when working as PD. The on/off ratio S (denoted as the ratio of photocurrent to dark current) is a crucial PD parameter, which can be used to evaluate its photoswitching performance [34]. A bias tee was used to filter out the DC current as the LED emitted light. To measure the photocurrent, a pulse signal was applied to the 405 nm LD, where the frequency of the pulse signal was 10 kHz and the duty cycle was 50%. A picoamp AC RMS ammeter was used to measure the RMS value of the AC current and calculate the corresponding peak value as the photocurrent. The dark current was measured directly using a DC source. The I–V curves of this point under darkness and a received optical power of 4.32 mW are shown in Fig. 5. For the mini-LED PD under zero-bias and reverse bias, the dark current after filtering the DC was 10^{-11} to 10^{-9} mA. After turning on, the dark current was approximately 10^{-8} to 10^{-7} mA because part of the current passed through the DC block. At a bias of 3 V, the peak value of the photocurrent was calculated to be approximately 0.56 mA, the corresponding dark current was 4.21×10^{-8} mA, and the corresponding S was up to 1.33×10^7 . The on/off ratio of the mini-LED PD when emitting light was higher than the previously reported values for GaN-based PDs [35].

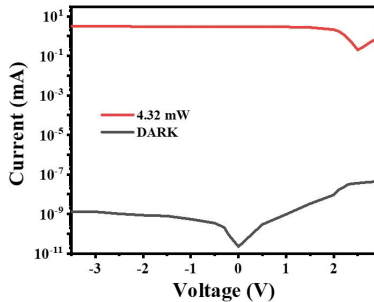


Fig. 5. I–V curve of the mini-LED PD in the dark and with a 4.32 mW input optical power after the DC current is filtered.

Furthermore, the specific detectivity (D^*) of the PD, which governs the signal-to-noise ratio, can be calculated as

$$D^* = R \times \sqrt{\frac{1}{2q \times J_d}} \quad (5)$$

where q is the charge constant, and J_d is the dark current density. D^* was calculated as 6.19×10^9 Jones at 3 V bias, which ensures the effective detection capability of signals from the mini-LED PD when emitting light.

Based on the aforementioned considerable optoelectronic performance, we further studied the frequency response of the mini-LED PD under different biases to verify the feasibility of turned-on mini-LED PDs as receivers for VLC. A vector network analyzer (VNA, Keysight E5071C) was used to output an AC frequency sweep signal combined with a DC signal through a bias-tee to drive the 405 nm LD, and after free space transmission, the signal was received by the mini-LED PD to measure the OE -3 dB bandwidth of mini-LED PD. The frequency responses under different bias are presented in Fig. 6(a). The calculated 3-dB bandwidth under different biases is shown in the inset. Similar to the responsivity, the bandwidth of the mini-LED PD decreased further when the bias voltage was larger than the turn-on voltage. This is also related to the fact that the built-in electric field was canceled by the forward voltage, reducing the speed of photogenerated carriers across the depletion layer, and resulting in an increase in the time for carriers to cross the depletion layer. In addition, the forward voltage reduced the width of the depletion layer. As the bias approached the turn-on voltage, the diffusion capacitance increased, which increased the device capacitance and reduced the photoresponse speed, resulting in a decrease in the bandwidth. The mini-LED PD had a bandwidth of 16 MHz at a bias of 3 V.

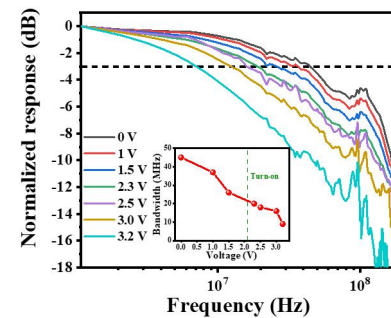


Fig. 6. (a) Frequency responses of the mini-LED PD when turned off and turned on.

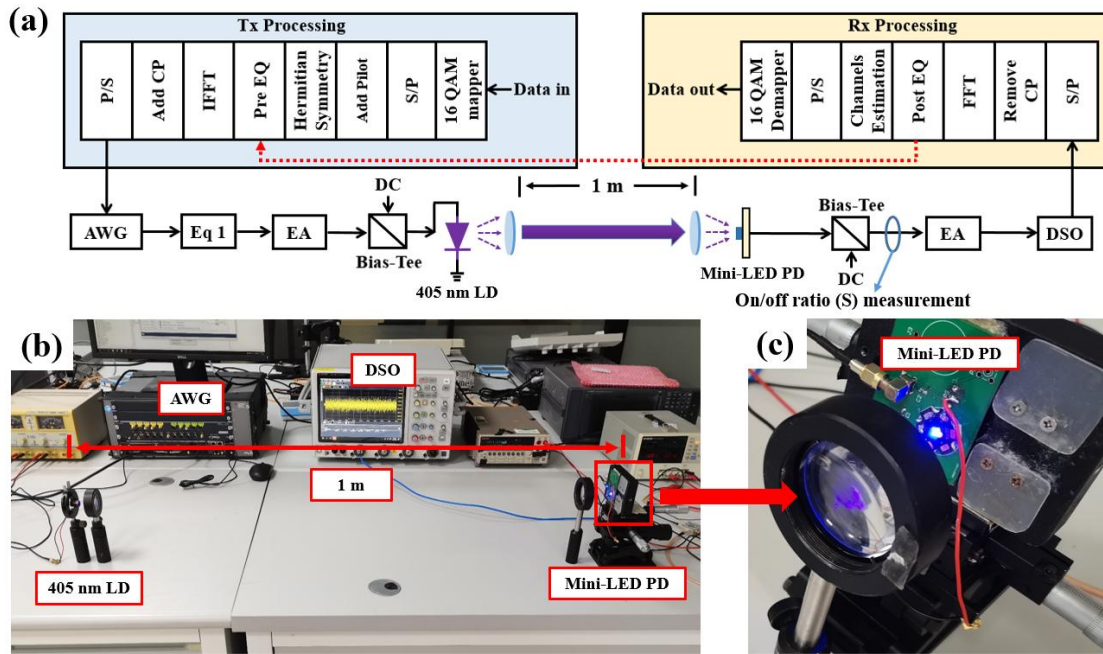


Fig. 7. (a) Schematic setup of the mini-LED PD-based VLC system. (b) The photograph of the mini-LED PD-based VLC system and (c) mini-LED working as a PD when turned on.

Figure 7(a) shows the experimental setup of the VLC system based on the mini-LED PD. Although the responsivity of the mini-LED PD gradually increases when the bias is greater than 2.5 V, excessive bias will lead to a decrease in the modulation bandwidth and an uneven frequency response. Considering the responsivity, bandwidth and output optical power of the mini-LED PD, in order to be used as the transmitter and receiver of the full-duplex VLC system simultaneously in subsequent experiments, we set the bias of the mini-LED PD as 3 V (corresponding to injection current of 11.5 mA). Additionally, to compensate for the high frequency signal attenuation, a cascaded bridged-T amplitude equalization circuit was used to improve the available signal bandwidth. The arbitrary waveform generator (AWG, Keysight M8190A) AWG is used to produce the electrical OFDM data to drive the 405 nm LD. The OFDM encoder at the Tx side contains symbol mapping to 16 quadrature-amplitude-modulation (QAM) mapper, input random data serial-to-parallel conversion (S/P), Add Pilot, Pre-Equalization (Pre-EQ), inverse fast-Fourier-transformed (IFFT), and cyclic-prefix insertion (CP), parallel-to-serial conversion (P/S). For the QAM OFDM data, the FFT size is set as 512. The number of subcarriers for data is 200. In addition, the numbers of CP and pilot tones are also fixed as 32 and 28, respectively. We changed the data rate mainly by adjusting the sampling rate of AWG.

The driving signal from AWG is equalized by the proposed amplitude equalizer. After amplified by electrical amplifier (EA, Mini circuits, ZX60-43-s+), the resulting signal coupled with direct current (DC) by bias-tee to drive the 405 nm LD. The optical signals emitted by the 405 nm LD were then received by the mini-LED PD after a 1 m free-space transmission distance. A source-measuring unit

(Keithley 2641) is used to provide drive voltage through the bias-Tee (Mini circuits, ZX85-12G-S+) after the mini-LED PD and monitor the photocurrent. The output signal of bias-Tee is amplified by EA (Mini circuits, ZX60-43-s+), and then fed into an oscilloscope (DSO, Keysight DSAV134A) and are acquired and transferred to a control computer for subsequent processing. Afterward, down-sampling, synchronizing, series-parallel-converting (S/P), removing CP, fast Fourier transform (FFT), parallel-series-converting (P/S), and 16 QAM demapper are implemented, sequentially. Before P/S, channels estimation and post equalization are employed. Iterative equalization processing is performed as shown in Fig. 7(a). After the channel estimation process, the received signal is post-equalized (post-EQ block) in the frequency domain by flattening the average amplitude for all subcarriers. Afterward, the value is sent to the pre-EQ block, and the subcarrier power of the pre-EQ block is readjusted.

Finally, the series data streams are demodulated and output. Fig. 7(b) and (c) shows the photograph of the experimental setup and the mini-LED operating as a PD when turned on.

The detailed circuit of proposed cascaded bridged-T amplitude equalization circuit is shown in Fig 8(a). The parameters of proposed cascaded bridged-T amplitude equalizer are $R_1 = 10 \text{ k}\Omega$, $R_2 = 100\Omega$, $R_3 = R_6 = R_7 = 50 \Omega$, $R_4 = R_8 = 20 \Omega$, $R_5 = 1 \text{ k}\Omega$, $C_1 = 50 \text{ pF}$, $C_2 = 100\text{pF}$, $C_3 = C_4 = 10 \text{ pF}$, $L_1 = L_2 = L_3 = L_4 = 22 \text{ nH}$. Figure 8(b) shows the improvement effect of the equalization circuit on the O-E modulation bandwidth of the mini-LED PD. The high frequency attenuation of the mini-LED PD at 3 V bias is serious without the equalization circuit, and the bandwidth is 16 MHz. With the equalization circuit, the frequency response is further flattened and the bandwidth is increased to 116 MHz.

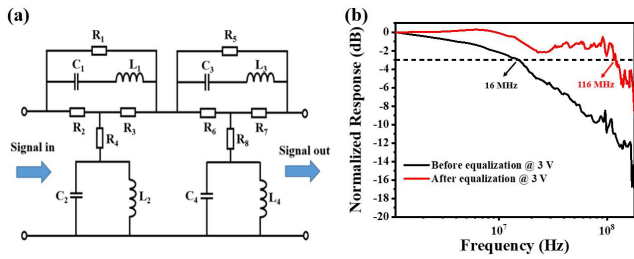


Fig. 8. (a) Two cascaded bridged-T amplitude equalizer circuit. (b) Improvement of the equalization circuit on the O-E modulation bandwidth of the mini-LED PD.

The performance of BER versus data rate of the VLC system was obtained, as shown in Fig. 9(a). The BER increases monotonically with the increase in data rate owing to the low response of the system to high frequency signals. The maximum data rate is 528 Mbps with BER of 2×10^{-3} below the forward error correction (FEC) limit of 3.8×10^{-3} . At the transmission rate of 528 Mbps, the OFDM signal occupies a bandwidth of 159.9 MHz. Even if the signal occupies a bandwidth slightly higher than the -3 dB bandwidth of the system, data transmission with a BER below the FEC limit can be achieved due to the channel

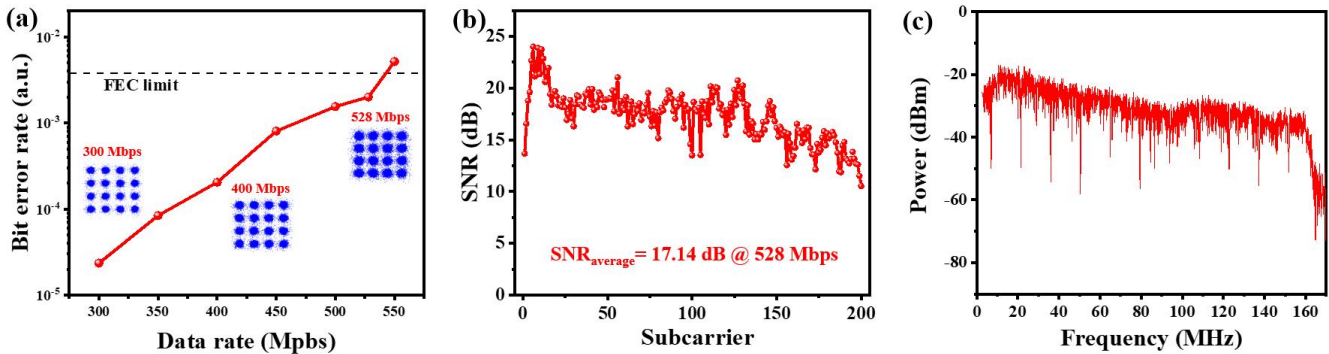


Fig. 9. (a) The performance of BER versus data rate and (c) the SNR distribution with every subcarrier. (d) The corresponding power spectra of 528 Mbps OFDM signals

III. FULL-DUPLEX VLC SYSTEM

A. Experiment setup

These results experimentally demonstrated that the mini-LED PD could simultaneously exhibit light emission and light detection behaviors, which has a propulsive role in the multifunction application of GaN-based LEDs. At the bias of 3 V, the measured light output power of the device was 4.1 mW, which makes it possible for the mini-LED PD to be used as a VLC Tx or display while also being used as a detector. The duplex VLC system based on this multifunctional device will simplify the structure and reduces power consumption.

To demonstrate this, we conducted a full-duplex VLC experiment with transmission distance of 1 m/1 m, the schematic of which is shown in Fig. 10(a). The AWG is used to generate electrical OFDM data to drive 405 nm LDs and mini-LED PDs. The encoding process of OFDM data is similar to the previous simplex link, the OFDM encoder at the Tx side contains symbol mapping to 16 QAM, S/P, Add Pilot, Pre EQ, IFFT, Adding CP and P/S. The setup for the uplink is similar to the previous experiment, using a 405 nm

estimation and equalization algorithm. It proves that the mini-LED can still be used as a detector for high-speed transmission even worked for illumination or display. BER is an important indicator of OFDM signal transmission performance, while signal-to-noise ratio (SNR) is the main factor that affects the BER of the communication system. A lower SNR means that the noise has a greater impact on the signal. For 16-QAM OFDM signals, the FEC criterion for SNR is 15.2 dB. The lower SNR corresponds to more dispersive constellation diagrams and correspondingly higher BERs. Figure 9(b) is the SNR distribution with every subcarrier. The average SNRs were 17.14 dB, which was sufficient to support the 16-QAM OFDM with a certain bandwidth. Based on the measurements of the subcarrier SNRs, we find that the SNRs of the high-frequency subcarriers are still attenuated even after the equalizer, due to the relatively low response to high-frequency signals in the VLC system. It leads to the fact that the bit errors occur more in the high-frequency subcarriers. The corresponding power spectra of 528 Mbps OFDM signals were captured, as shown in the Fig. 9(c).

LD as the transmitter. After a 1 m free-space transmission, the mini-LED PD is used for opto-electrical (O/E) conversion with a post-stage bias-Tee and fed into the oscilloscope after being amplified by EA.

At the same time, the downlink OFDM data is loaded onto the mini-LED PD, making it act as a downlink transmitter. The light emitted by the mini-LED PD was focused by the other two lenses and received by an APD (APD 500, 1.25G), acting as the downlink receiver. A blue filter was used to remove light reflected by the laser during the experiment. A power amplifier was connected to the APD to further amplify the voltage. The received signals from mini-LED PD and APD are routed to an oscilloscope and are acquired and transferred to a control computer for subsequent processing. Afterward, down-sampling, S/P, removing CP, FFT, channels estimation, signals separation, post equalization, P/S, and 16 QAM demapper are implemented, sequentially. In the full-duplex VLC system described above, the single mini-LED acted as both the uplink receiver and downlink transmitter. Figs. 10(b)–(d) show photographs of the experimental setup, mini-LED PD, and APD used as the downlink receiver, respectively.

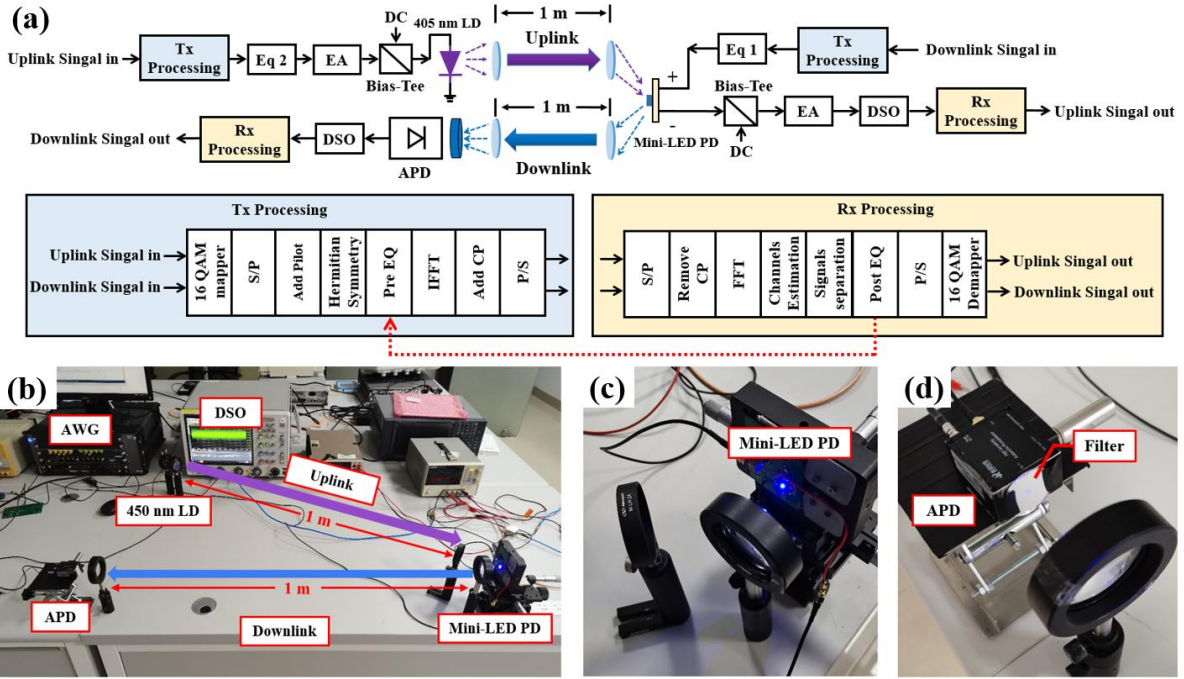


Fig. 10. (a) Schematic setup and (b) photograph of the mini-LED PD-based full duplex VLC system. (c) Mini-LED working as both the uplink receiver and downlink transmitter, and (d) APD used as the downlink receiver.

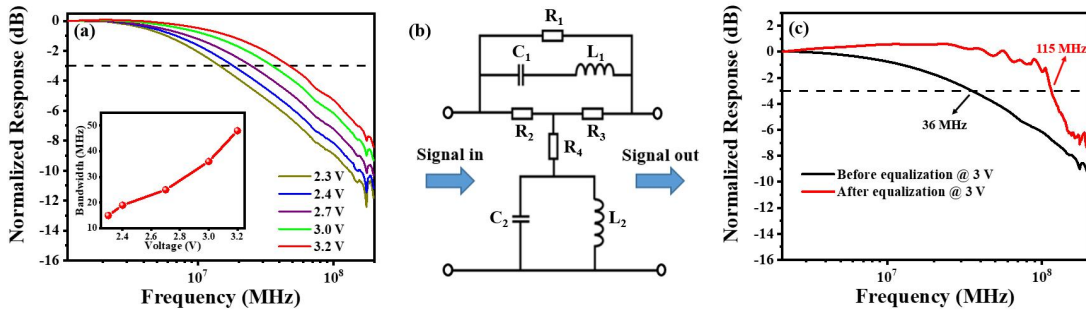


Fig. 11. (a) Frequency responses of the mini-LED when used as a transmitter under different bias voltages. (b) Bridged-T amplitude equalizer circuit. (c) Improvement of the equalization circuit on the modulation bandwidth of the mini-LED when used as a transmitter.

B. Results and discussions

The emission frequency response of the mini-LED PD at different bias from 2.3 to 3.2 V is shown in Fig. 11(a). The modulation bandwidth was 36 MHz at a bias of 3 V, as shown in the inset. In order to further improve the transmission rate, another bridged-T amplitude equalizer (Eq 2) is also designed for mini-LED as VLC Tx. The detailed circuit of Eq 2 are shown in Fig. 11(b). The parameters used for proposed bridged-T amplitude equalizer are $R_1 = 10 \text{ k}\Omega$, $R_2 = R_3 = R_4 = 50 \text{ }\Omega$, $C_1 = C_2 = 50 \text{ pF}$, $L_1 = 18 \text{ nH}$, $L_2 = 22 \text{ nH}$. Under a bias voltage of 3 V, the emission bandwidth of the mini-LED can be increased to 115 MHz, as shown in Fig. 11(c).

It is worth noting that as the LEDs transmit and receive data simultaneously, the output current will contain two signal components, the photocurrent responded from the received signal and the current from the driving signal. This is the same for the APD at the downlink receiver, because

the received signal from the mini-LED also contains both components. Hence, in this system, the upstream and downstream receivers behave similarly to the receivers of Multiple Input Single Output (MISO) VLC systems. After the channel estimation, signals separation was need to implement to extract the valid signals of the uplink and downlink.

First, we analyze the signal output by the mini-LED PD. As the mini-LED PD transmit and receive data simultaneously, the output signal will contain two signal components. Therefore, for this full duplex system, the uplink can be regarded as a 2×1 MISO VLC system. The input signals are from 405 nm LD and the AC signal from the AWG. The received signal of the mini-LED PD can be regarded as a mixed data of the two streams after passing through different channels. In the Rx Processing shown in Fig 10 (a), the channels of the two streams need to be analyzed and separated separately.

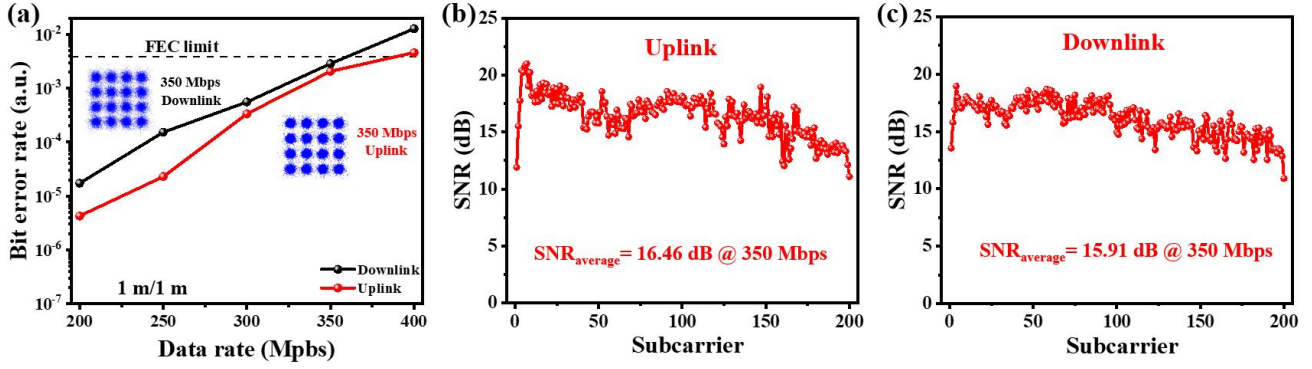


Fig. 12. (a) The BER performance with different transmission rate of the uplink and downlink of the full-duplex VLC system. (b) the SNR distribution with every subcarrier for the (b) downlink and (c) uplink.

Hence, at the receiving end, we use the traditional MISO OFDM VLC system processing method to separate the uplink and downlink OFDM signals.

Taking the uplink as an example to illustrate the signal separation algorithm, The MISO channel can be modelled as

$$\mathbf{y}(t) = G_{TIA} \times (\mathbf{H} \times \mathbf{x}(t) + \mathbf{n}(t)) \quad (6)$$

where $\mathbf{y}(t)$ is the voltage signal vector output of the uplink; G_{TIA} is the current-voltage conversion gain of the bias-tee and Amp; $\mathbf{H}$ is a 2-by-1 channel matrix; $\mathbf{x}(t)$ is a 2-by-1 transmitted signal vector, broadcast by the 2 Tx from the photocurrent of mini-LED PD, and the downlink signal applied directly to the mini-LED PD; and $\mathbf{n}$ is the 2-by-1 channel noise vector. To estimate the channel matrix $\mathbf{H}$ for demultiplexing, each Tx is enabled in turn, then insert the training sequence $\mathbf{x}_i(t)$ into the uplink and downlink OFDM signals and send to Tx and the stream $\mathbf{y}_i(t)$ collected from the Rx. The element h_i of $\mathbf{H}$ corresponding to this particular link is estimated from

$$h_i^2 = \frac{\sum_{t=0}^T \mathbf{y}_i^2(t) - \sigma^2 T}{\sum_{t=0}^T \mathbf{x}_i^2(t)} \quad (7)$$

where T representing the duration of the pilot training wave and σ^2 is the noise variance of the Rx.

OFDM signals separation can be achieved by pre-multiplying $\mathbf{y}(t)$ with a matrix $\mathbf{G}$, such that $\mathbf{z}(t)$ defined as follows is considered an approximation of $\mathbf{x}(t)$:

$$\mathbf{z}(t) = \mathbf{G} \times \mathbf{y}(t) \quad (8)$$

As a compromise between complexity and performance, MMSE demultiplexing is used in our demonstration. During the working process of the system, we adjust the amplitude of the uplink and downlink signals so that the average transmission power ρ of the uplink and downlink OFDM signal components in the output superimposed signal is approximately equal. The matrix $\mathbf{G}$ thus takes the form:

$$\mathbf{G} = \frac{\mathbf{H}^\dagger}{\mathbf{H}\mathbf{H}^\dagger + \frac{\mathbf{C}_N}{\rho}} \quad (9)$$

where $\mathbf{H}^\dagger$ is the Hermitian transpose of $\mathbf{H}$; $\mathbf{C}_N$ is the ensemble covariance matrix of $\mathbf{n}(t)$. The vector $\mathbf{z}(t)$ defined in formula (8) is then passed for subsequent processing. Extract the uplink valid signal from $\mathbf{z}(t)$. After signal separation is completed, iterative equalization processing is also performed.

The separation and extraction process of signal in the downlink is similar to that of the uplink. It should be noted that the pre-equalization and post-equalization in the system are carried out independently on the uplink and downlink.

After separating the uplink and downlink signals from the outputs of the mini-LED PD and APD, respectively, the BER performance with different transmission rate of the uplink and downlink of the full-duplex VLC system is analyzed, as shown in Fig. 12(a). Considering that the uplink and downlink bandwidths are not much different after equalization, the data transmission rates of the uplink and downlink are the same during the test. With the transmission rate of 350 Mbps, the BER of uplink and downlink are 2.1×10^{-3} and 2.9×10^{-3} , respectively, which are both below the FEC limit. The inset of Fig 12(a) presents the constellation diagrams of the received 16QAM-OFDM signals received in the uplink and downlink. We further measured the SNR of the uplink and downlink signals as a function of subcarrier at a transmission rate of 350 Mbps, as shown in Fig. 12(b) and (c). The average SNR of the uplink and downlink subcarriers are 16.46 and 15.91 dB, respectively. Compared with the SNR from the mini-LED PD-based simplex communication at the same rate, the BERs and SNRs of the subcarriers in the full-duplex link are reduced. In addition to the decrease in the SNR caused by the algorithm separating the signals, it is also possible that the downlink signal increased the bias of the mini-LED PD by transmitting at a high level, limiting the bandwidth.

To measure the performance of this full-duplex communication system at longer transmission distances, we extended the transmission distances to 1.2 m/1.2 m and 1.5 m/1.5 m, corresponding BER performances are shown in Fig. 13 (a) and (b). We can find that the BERs of the uplink does not change significantly with the transmission distance increases, while that of the downlink increases obviously. The reason is that the light emitted by the mini-LED has a greater power loss during transmission. At a distance of 1.2 m/1.2 m, a transmission rate of 300 Mbps for the uplink and downlink was achieved, with BERs of 5.3×10^{-4} and 2.9×10^{-3} . Similarly, at a distance of 1.5 m/1.5 m, both the uplink and downlink are capable of 225 Mbps transmission rates, with BERs of 2×10^{-5} and 3.3×10^{-3} .

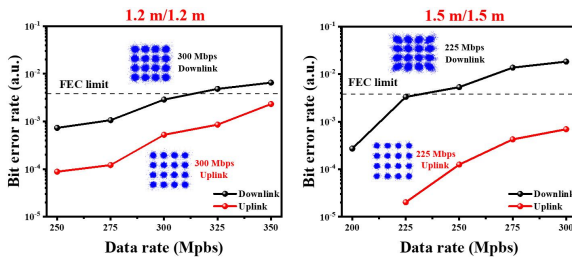


Fig. 13. The BER performance of the full-duplex VLC system at a transmission distance of (a) 1.2 m/1.2 m and (b) 1.5 m/1.5 m.

Based on the above discussion, a single mini-LED works as the receiver of the uplink and the transmitter of the downlink simultaneously to realize full-duplex VLC of 350 Mbps at transmission distance of 1 m/1 m, which can realize the function similar to the antenna in wireless communication. In addition to serving as the Tx and Rx simultaneously, a single mini-LED PD can also provide display functions, further enhancing the versatility of LEDs. With the distance increased to 1.5 m/1.5 m, the system is still capable of full-duplex transmission of 225 Mbps. Considering the relatively low injection current density of 120 A/cm², this transmission distance is already 17 considerable compared to other VLC systems working under low injection [18]. In the future, MIMO VLC systems can be implemented using LED arrays to further increase the transmission rate.

IV. CONCLUSION

In conclusion, we investigated the performance of an 80 × 120 μm² blue mini-LED as a PD when emitting light. An Al₂O₃ passivation was deposited on the sidewalls of the device using ALD. In addition to improving the external quantum efficiency of mini-LEDs and reducing leakage current, sidewall passivation also improves the responsivity of the mini-LED PD from 0.209 A/W to 0.243 A/W at a forward bias voltage of 3 V. A equalizer circuit was used to extend the OE -3dB bandwidth of mini-LED PD from 16 MHz to 116 MHz at 3 V bias. With the mini-LED acting as PD, a VLC link over 1 m with a data rate of 528 Mbps is achieved using 16 QAM-OFDM. This indicates that the mini-LED could still act as PD for VLC, even at an injection current sufficient for display. In addition, we further designed a full-duplex communication system in which a single mini-LED biased at 3 V acted as both an uplink detector and a downlink transmitter. With transmission distance of 1m/ 1m, a transmission rate of 350 Mbps for the uplink and downlink was achieved. With transmission distance increasing to 1.2 m/1.2 m and 1.5 m/1.5 m, full-duplex VLC systems are also capable of achieving transmission rates of 300 Mbps and 225 Mbps, respectively. The proposed method allows the same mini-LED to be used simultaneously as a display, VLC Tx, and VLC Rx at relatively low operating current density, which simplifies the structure of the full-duplex communication system, reduces system power consumption, facilitates the integration of multifunctional small LED devices, and verifies the potential of mini-LED applications in the field of miniaturized IoT and multi-functional display.

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